

Reducing False Alarms in Edge Driver Monitoring Systems through Negative Frame Curation and Hard Negative Mining

Author names are omitted for double-blind review

Abstract—Frequent false alarms in edge-deployed driver monitoring systems degrade driver trust, waste on-device computing cycles, and elevate thermal load. This paper evaluates whether curating negative training frames reduces false alarms and whether replacing randomly sampled negative frames with hard-mined false positives provides further suppression. Four lightweight You Only Look Once (YOLO) object detection models were trained and benchmarked on a 15,723-frame dataset extracted from the Driver Monitoring Dataset across four localized visual cues using a dedicated graphics processing unit. Models were evaluated across negative-frame ratios ranging from 0% to 80%, followed by hard-negative mining trials and an execution latency benchmark measuring per-frame runtime against standard 30 ms real-time in-cabin deadlines, providing an edge-proxy compute baseline prior to low-power embedded deployment. Results demonstrate that increasing negative exposure consistently reduces false positives across all evaluated detectors, achieving a 6.4× to 25.7× reduction at the 80% configuration while maintaining stable detection performance. At the 40% reference ratio, hard negative mining provides an additional 33.3% to 67.8% reduction in false positives compared with random negative sampling, with the magnitude of improvement varying across models. Runtime profiling demonstrates sub 30 ms inference latency under batch size one execution without modifying model architecture or inference complexity. These results demonstrate that negative frame curation and hard negative mining provide an effective and computationally inexpensive approach for improving the operational reliability of lightweight edge Driver Monitoring Systems.

Index Terms—Driver monitoring systems, object detection, false alarm suppression, hard-negative mining, edge computing.

I. INTRODUCTION

Beginning 7 July 2024, European Union (EU) regulations required Advanced Driver Distraction Warning (ADDW) systems in newly type-approved vehicles, and this mandate extends to all newly registered vehicles from 7 July 2026 [1]. Early vision-based Driver Monitoring Systems (DMS) estimated driver states using spatio-temporal action recognition [2], [8]. However, these three-dimensional convolutional

This study involves secondary analysis of human subject data. The authors confirm that all experimental protocols are exempt from institutional review board oversight (under category 45 CFR 46.104(d)(4) for publicly available secondary research data). All facial image excerpts reuse data from the open-access Driver Monitoring Dataset (DMD) under its non-commercial research distribution license, with participant consent obtained by the original dataset authors. In alignment with applicable data protection frameworks (including GDPR Art. 6(1)(f) and Art. 9(2)(j) for scientific research), facial data were evaluated strictly as non-commercial secondary research on consented public benchmarks.

architectures require excessive memory and compute time, making them unsuitable for low-power automotive micro-controllers [3], [8]. Modern edge DMS pipelines instead formulate monitoring as frame-by-frame object detection [9], [11]. These pipelines deploy lightweight models such as the You Only Look Once (YOLO) detector family to identify localized behavioral cues, including phone use, drinking, and eye closure [2], [10]. In-cabin edge hardware must run these models under strict limits: inference latency must remain under 30 ms [4], and power draw must stay below 10–15 W to avoid overheating [3], [6].

Lightweight detectors meet these power and latency targets, but existing DMS datasets consist almost entirely of positive distraction events [9], [11]. Because single-frame detectors lack temporal context, models trained without negative examples generate frequent false positives on benign objects like steering wheel contours, seat belts, and clothing folds [8]. Naturalistic on-road studies indicate that potentially distracting secondary-task engagement occupies a small share of total driving time—one widely cited video-based study measured such engagement, excluding passenger conversation, at just 14.5% of total driving time [18]—so cue-free driving constitutes the substantial majority of on-road operating time. Our curated evaluation stream standardizes background prevalence at $p_{\text{neg}} = 0.809$ (80.9% negative frames, matching our held-out test split) to benchmark false-alarm suppression under controlled in-cabin conditions. Frequent false alarms during normal driving cause alert fatigue, leading drivers to ignore, mute, or bypass safety systems entirely [5], [13]. These false alarms also trigger downstream tasks, including facial landmark tracking, Controller Area Network (CAN) bus logging, and video recording [3], [4]. These secondary tasks create processing spikes that cause thermal throttling on embedded chips [6]. Adding negative (background-only) frames during training reduces false alarms, but the impact of negative-sampling ratios and hard-negative mining has not been thoroughly evaluated in edge DMS applications [16], [17].

To address these challenges, this paper provides three main contributions:

- **Negative Sampling Benchmarks:** A systematic evaluation of negative-to-positive frame ratios (0% to 80%) across four lightweight edge-oriented object detectors (YOLO11n, YOLO26n, YOLO12n, and YOLOv10n)

[10], [12].

- **Hard-Negative Mining Analysis:** An experimental evaluation showing that targeted hard-negative mining lowers the false alarm rate (FAR) by up to 67.8% without reducing cue detection accuracy [14].
- **Operational Nuisance Alert Modeling:** An operational formulation bridging frame-level false alarm rates to hourly nuisance alert rates ($\mathcal{A}_h$) across streaming regimes (5, 15, and 30 FPS), paired with real-time latency profiling against the 30 ms in-cabin deadline to evaluate operational reliability and cascading edge thermal overheads.

Specifically, this study answers two central research questions (RQs):

- **RQ1:** What ratio of negative-to-positive training frames minimizes false alarms without reducing cue detection sensitivity, and is this ratio consistent across different lightweight architectures?
- **RQ2:** Does replacing random negative frames with hard negatives (prior false positives) provide better false-alarm suppression in resource-constrained models?

II. RELATED WORK

A. Vision-Based Driver Monitoring and Cue Detection

Vision-based DMS historically formulated driver state estimation as action recognition [2], [8]. Early methods used convolutional neural networks (CNNs) to classify video clips into coarse categories, such as safe driving, talking, or drinking [2], [8]. Multimodal benchmarks, including the Driver Monitoring Dataset (DMD) [2] and Drive&Act [7], established standardized evaluation protocols across red-green-blue (RGB), depth, and infrared camera streams. However, 3D CNNs and long short-term memory (LSTM) networks impose latency and memory requirements that exceed automotive microcontroller limits [3], [8].

To satisfy real-time constraints, modern edge DMS pipelines reframe monitoring as frame-by-frame object detection using lightweight YOLO models [9], [11]. Recent works propose customized architectures: YOLO-Drive incorporates state-space blocks and attention mechanisms [9], P-YOLOv8 leverages a pretrained YOLOv8 backbone with a lightweight classification head optimized for TinyML deployment [11], and Chaman *et al.* benchmark compact YOLOv12 models [10]. Yet, these detectors are trained almost entirely on positive distraction intervals [9], [11]. Without background regularization, single-frame models lack temporal context and frequently misclassify steering wheels, seat belts, and clothing folds as distraction cues [8]. Rather than designing bespoke architectures, this work introduces a data-centric evaluation of negative-frame curation and hard-negative mining to suppress false alarms across standard YOLO detectors.

B. Automotive Edge Computing and Cascading Thermal Overhead

Automotive edge platforms must maintain sub-30 ms inference within a 10–15 W power budget under passive cooling [3], [4]. Embedded DMS studies typically evaluate frame

rates and nominal power under static workloads [3]. In deployed automotive pipelines, however, cue detections trigger secondary tasks, including facial landmark alignment, gaze tracking, CAN bus logging, and video ring-buffer encoding [3], [4].

When an unregularized detector produces frequent false alarms, these secondary tasks execute repeatedly. Benoit-Cattin *et al.* demonstrated that sustained processing bursts cause thermal throttling in embedded vision processors, triggering dynamic frequency scaling that reduces inference throughput by over 40% [6]. Existing DMS benchmarks evaluate detectors in isolation, overlooking how frame-level false positives compound into operational hourly nuisance alerts ($\mathcal{A}_h$) that induce thermal stress.

C. Alert Fatigue, Automation Disuse, and Regulatory Mandates

European Union (EU) vehicle-safety regulations require reliable distraction detection and prompt warning escalation [1]. However, persistent false alarms compromise safety system efficacy. In human-automation interaction theory, Parasuraman and Riley established that false alarms are the primary cause of automation *disuse*—the intentional muting, ignoring, or disabling of safety systems by operators [13]. Repeated false alerts during routine driving induce alert fatigue and erode driver trust, motivating motorists to circumvent monitoring hardware with aftermarket defeat devices [5], [13]. Suppressing false alarms is therefore essential for driver acceptance and regulatory compliance.

D. Negative-Frame Curation and Hard-Negative Mining

In object detection, false-positive suppression typically relies on loss reweighting, such as Focal Loss [15], or candidate resampling, such as Online Hard Example Mining (OHEM) [14]. However, online anchor-level mining like OHEM is computationally impractical for lightweight single-stage architectures, and loss reweighting cannot compensate when training sets lack background negative frames entirely [15]. Instead, offline image-level bootstrapping—harvesting false-positive frames from unannotated streams and augmenting subsequent training passes—provides a computationally viable alternative for compact detectors.

Practitioner heuristics often recommend adding 0% to 10% background frames during training [16]. Yet, this guideline was designed for general object detection benchmarks and remains unvalidated in automotive environments, where naturalistic on-road evidence indicates that potentially distracting secondary-task engagement occupies only a small share of total driving time [18], yet standard object-detection benchmarks under-sample these baseline, cue-free periods. Furthermore, uniform random negative sampling yields diminishing returns because simple background scenes generate minimal gradient feedback [14], [17]. While video-based mining techniques harvest temporal variations as hard negatives in generic domains [17], prior works have not systematically evaluated negative ratios from 0% to 80% across lightweight YOLO detectors or

measured the impact of hard-negative mining on operational nuisance alert rates ($\mathcal{A}_h$).

III. METHODOLOGY

This section details the models, dataset, split protocols, metrics, and experimental procedures designed to address two central research questions:

- **RQ1 (Negative Sampling Ratio):** What ratio of negative-to-positive training frames minimizes background false alarms without degrading cue detection sensitivity, and is this threshold consistent across lightweight architectures?
- **RQ2 (Hard-Negative Mining):** Does replacing random background frames with hard negatives (prior false positives) provide superior false-alarm suppression at matched training dataset size?

Sections III-A–III-D define the architectures, dataset curation, split protocols, and evaluation metrics for RQ1. Sections III-E–III-F detail the hard-negative mining protocol and edge deployment benchmarks for RQ2.

A. Model Architectures and Training Configuration

We benchmark four lightweight You Only Look Once (YOLO) detectors containing under 3 million parameters: YOLO11n [20], YOLO26n [22], YOLO12n [21], and YOLOv10n [19]. Table I summarizes each model’s parameter count, computational complexity in giga floating-point operations (GFLOPs), and input resolution.

TABLE I
EVALUATED YOLO ARCHITECTURES

Model	Params (M)	GFLOPs	Input Size
YOLO11n	2.6	6.5	640×640
YOLO26n	2.6	6.2	640×640
YOLO12n	2.6	7.6 [†]	640×640
YOLOv10n	2.3	6.6	640×640

[†] Profiled via `thop` at FP32; vendor specification reports 6.5 GFLOPs.

TABLE II
TRAINING HARDWARE AND HYPERPARAMETERS

Parameter	Value
GPU	NVIDIA RTX 4060 (8 GB VRAM)
Pretrained weights	COCO
Epochs	100
Batch size	16
Optimizer	AdamW
Initial LR	0.00125
Weight decay	0.0005
Warmup epochs	3.0
LR schedule	Linear decay
Input size	640×640

All models are initialized from Common Objects in Context (COCO) pretrained checkpoints and trained via Ultralytics using the configuration in Table II on an NVIDIA RTX 4060 graphics processing unit (GPU) with 8 GB of video

RAM (VRAM). Dataloader workers and PyTorch random generators were initialized with the designated run seed ($\text{seed} \in \{42, 43\}$) under `deterministic=True` to ensure reproducible batch sequencing. Data augmentation includes horizontal flipping ($p = 0.5$), hue-saturation-value (HSV) shifts, translation ($\pm 10\%$), random erasing ($p = 0.4$), and mosaic augmentation (disabled during the final 10 epochs via `close_mosaic=10`). Geometric distortions (rotation, shear, and perspective warp) are omitted to preserve in-cabin camera geometry.

B. Dataset Curation and Annotation Protocol

We extract 15,723 face-crop frames at 1 frame per second (FPS) from 81 red-green-blue (RGB) recordings across 14 drivers in the Driver Monitoring Dataset (DMD) [2]. The curated set contains 3,001 positive cue frames (19.1%) and 12,722 negative frames (80.9%), establishing a background-heavy test stream that approximates the operational rarity of distraction events. Negative frames contain no bounding boxes. Positive frames contain exactly one localized bounding box identifying one of four driver cues: yawning, hand-over-mouth, drinking, or phone use (Fig. 1(a)).

Annotations were generated by a single annotator under three rules:

- **Mutual Exclusion:** If a driver covers a yawn with a hand, the behavior is labeled exclusively as hand-over-mouth.
- **Visible Anatomy Only:** Bounding boxes enclose only directly visible cue interactions without extrapolating occluded joints.
- **Scale Invariance:** No minimum bounding-box pixel dimension is imposed at the 640×640 resolution.

C. Training Splits, Negative Ratios, and Seed Protocol

We apply an 80/10/10 stratified split across the 3,001 positive frames, allocating 2,401 positive frames to training and 300 each to validation and testing.

TABLE III
DATASET SPLITS AND FRAME DISTRIBUTIONS

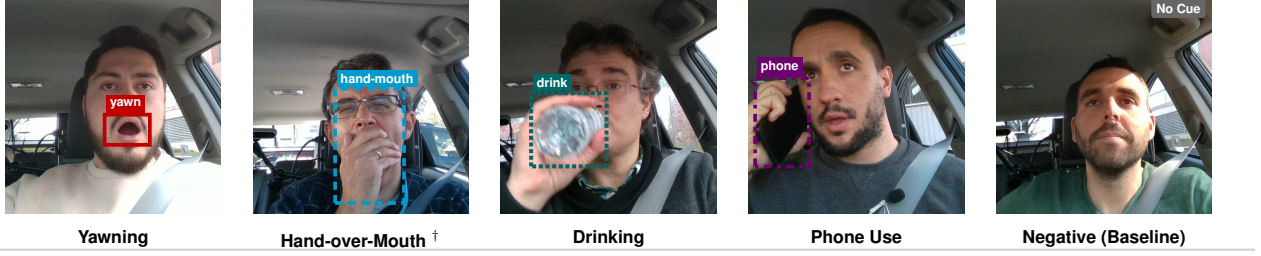
Split	Frame Counts		
	Positive	Negative	Total
0%	2,401	0	2,401
20%	2,401	600	3,001
40%	2,401	1,600	4,001
60%	2,401	3,602	6,003
80%	2,401	9,604	12,005
Val [†]	300	1,272	1,572
Test [†]	300	1,272	1,572

[†] Held-out evaluation splits feature an 80.9% negative ratio.

TABLE IV
RANDOM SEED PROTOCOL BY NEGATIVE RATIO

Negative Ratio	Seeds Used
0%, 20%, 40%	42, 43
60%, 80%	42

(a) Localized Behavioral Cue Annotations vs. Baseline Negative



(b) Training Pipeline and Closed-Loop Hard-Negative Mining



Fig. 1. Overview of the training and deployment pipeline: (a) Target behavioral cues (640×640 pixels) with an unannotated negative frame (baseline driving). When a driver covers a yawn, the action is labeled as hand-over-mouth ($\dagger$). Line styles and colors provide dual encoding for greyscale clarity. (b) The training pipeline evaluates negative-frame ratios and harvests false positives (FPs) to retrain the model, reducing projected nuisance alert rates ($\mathcal{A}_h$) for in-cabin driver monitoring systems (DMS); the final stage reflects desktop-GPU latency profiling used as a compute proxy, not validated automotive SoC hardware.

For RQ1, we construct five training configurations by augmenting the 2,401 positive frames with varying negative-frame proportions from 0% to 80% (Table III). Both the validation and test sets maintain a fixed 80.9% negative ratio (1,272 negative and 300 positive frames, 1,572 total each) to benchmark detector false-positive resistance under a standardized background-heavy distribution. To account for initialization and batch-shuffling variance, configurations under sparse negative supervision (0%, 20%, and 40%) were trained across two random seeds ($\text{seed} \in \{42, 43\}$) and reported as dual-seed means. Due to training compute budgets across the 20-configuration matrix, 60% and 80% sweeps were trained under a single baseline seed ($\text{seed} = 42$), as listed in Table IV. While the monotonic reduction across all four architectures corroborates the stability of the 80% ratio, future evaluations will extend multi-seed coverage uniformly across all ratios.

D. Evaluation Metrics and False-Alarm Quantification

Models are evaluated across detection precision, recall, F1 score, mean Average Precision (mAP), false alarm rate (FAR), and false positives per 1,000 frames (FP/1k). Detections undergo non-maximum suppression (NMS) using an IoU threshold of 0.70 and an operational confidence threshold of $\tau = 0.25$. We note that while false alarms on a positive-only model can be partially reduced by simply raising τ post-hoc, doing so severely degrades foreground recall for subtle behavioral cues (such as partial eye closure or occluded phone use); background negative regularization is therefore required to suppress false alarms at fixed operational sensitivity.

A prediction is scored as a True Positive (TP) if it predicts the correct cue class with an Intersection over Union (IoU) ≥ 0.50 against a ground-truth box; otherwise, it is a False Positive (FP). Ground-truth boxes with no matching detection are False Negatives (FN). Precision, recall, and optimal F1 score are defined as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (1)$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (2)$$

Localization quality is measured using mAP_{50} across the $C = 4$ driver-cue classes:

$$\text{mAP}_{50} = \frac{1}{C} \sum_{c=1}^C \text{AP}_c^{50}, \quad (3)$$

where AP_c^{50} is the Average Precision for class c at an IoU threshold of 0.50. We also report $\text{mAP}_{50:95}$, averaged across ten IoU thresholds from 0.50 to 0.95 in 0.05 increments.

To quantify false alerts on background scenes, let FP_{neg} denote detections with confidence $\geq \tau$ ($\tau = 0.25$) produced on the $N_{\text{neg}} = 1,272$ unannotated test frames ($\mathcal{D}_{\text{eval}}^{\text{neg}}$). Per-frame FAR and scaled operational FP/1k are defined as:

$$\text{FAR} = \frac{FP_{\text{neg}}}{N_{\text{neg}}}, \quad \text{FP/1k} = 1,000 \times \text{FAR}. \quad (4)$$

E. Hard-Negative Mining Protocol

To address RQ2, we implement the closed-loop hard-negative mining protocol shown in Fig. 1(b). We first train

a baseline model f_{θ_0} using the 0% negative baseline (2,401 positive frames). Lacking background regularization, f_{θ_0} readily triggers on benign cabin features, serving as an effective filter for discovering informative non-target patterns.

We run f_{θ_0} in batched inference over unselected training background frames ($\mathcal{U}_{\text{neg}}$, strictly isolated from validation and test sets). For frame $X \in \mathcal{U}_{\text{neg}}$, let $\text{conf}(X)$ denote the maximum detection confidence. Frames triggering false detections with $\text{conf}(X) \geq \tau$ ($\tau = 0.25$) are harvested as hard negatives:

$$\mathcal{H}_{\text{mined}} = \{X \in \mathcal{U}_{\text{neg}} \mid \text{conf}(X) \geq \tau\}. \quad (5)$$

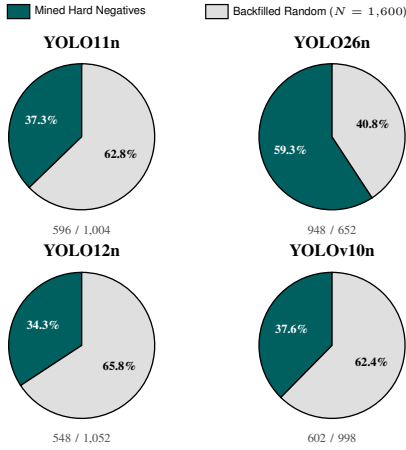


Fig. 2. Proportion of mined false positives versus backfilled random negatives per model (40% curation ratio, total $N = 1,600$). Subtitles report exact mined / backfilled frame counts.

Because the harvested hard-negative count varies across architectures (Fig. 2), we fix the total negative quota to $N_{\text{target}} = 1,600$ to match the 40% reference curation ratio (4,001 total frames, Table III). Frames in $\mathcal{H}_{\text{mined}}$ are ranked by descending confidence, and remaining slots are backfilled via uniform random sampling (seed = 42) from $\mathcal{U}_{\text{neg}} \setminus \mathcal{H}_{\text{mined}}$. Both the RQ1 40% random negative set and the candidate pool $\mathcal{U}_{\text{neg}}$ for RQ2 mining were drawn from the identical 10,178 training background partition, ensuring that random and hard-mined configurations share the same baseline distribution while isolating the effect of false-positive sample selection. Holding dataset cardinality constant isolates sample information content from training set volume. Detectors are retrained from COCO checkpoints across two seeds (seed $\in \{42, 43\}$) using the baseline training protocol.

F. Real-Time Deployment and Runtime Overhead Evaluation

To quantify operational impact, we project empirical FAR into expected hourly nuisance alerts, $\mathcal{A}_h$. For camera streaming rate r FPS, one hour of driving (3,600 s), and an evaluation background driving prevalence of $p_{\text{neg}} \approx 0.809$, assuming independent per-frame inference:

$$\mathcal{A}_h = 3600 \times r \times p_{\text{neg}} \times \text{FAR}. \quad (6)$$

While real-world false detections from static cabin features (such as seatbelt brackets or steering wheel seams) often occur in temporally correlated bursts rather than independent Bernoulli trials, (6) provides an unmitigated upper-bound rate for comparing raw detector false-alarm resistance. We evaluate $\mathcal{A}_h$ across three streaming rates: $r \in \{5, 15, 30\}$ FPS, corresponding to low-power sampling, typical embedded vision capture, and full real-time streaming.

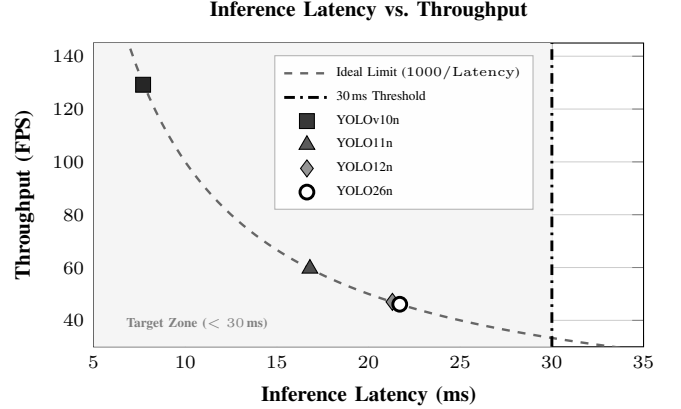


Fig. 3. Inference latency versus throughput across evaluated YOLO models. Each model uses a distinct marker shape for clarity in greyscale printouts. The dashed vertical line denotes the 30 ms real-time deadline.

Runtime efficiency is profiled on an NVIDIA RTX 4060 GPU using single-image batches (640×640 , FP32 precision) averaged over 200 iterations after 50 warm-up runs. While dedicated automotive system-on-chip (SoC) hardware with passive thermal limits (10–15 W) represents the ultimate target deployment, this GPU benchmark establishes an execution latency baseline. As shown in Fig. 3, all four models execute well within the 30 ms real-time deadline (7.7–21.7 ms per frame, achieving 46.1–129.2 FPS) with peak memory consumption under 1.5 GB of VRAM, confirming that architectural complexity does not violate frame-rate requirements.

IV. EXPERIMENTAL RESULTS

A. Negative-to-Positive Ratio Sweep

To address RQ1, we evaluate each detector across five negative-to-positive training ratios: 0%, 20%, 40%, 60%, and 80% (0 : 1 to 4.00 : 1; Table III). Fig. 4 presents detection accuracy and false-positive rates on the held-out test set (1,272 negative frames).

Convergence at High Ratios: Detectors trained without negative frames (0%) produce high false-alarm rates of 45.20 to 80.59 FP/1k (Fig. 4). Increasing negative frames to 80% reduces false alarms across all models to a narrow range of 3.14 to 7.08 FP/1k, representing a $6.4\times$ to $25.7\times$ reduction. Across all 20 evaluated configurations, the negative ratio strongly correlates inversely with FP/1k (Spearman $\rho = -0.937$, $p < 10^{-4}$). We note that these 20 points are not fully independent observations, since each of the four architectures contributes

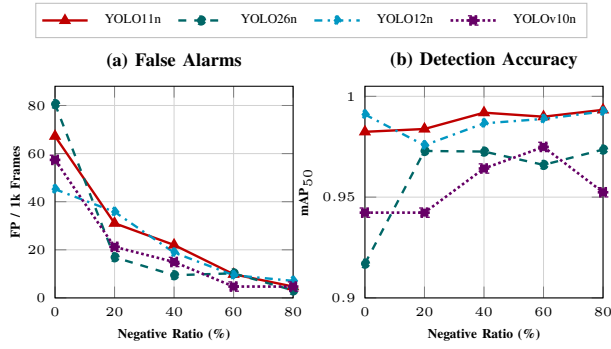


Fig. 4. Impact of negative-to-positive curation ratios (RQ1): (a) false alarms per 1,000 frames (FP/1k) across curation ratios; (b) core detection accuracy (mAP₅₀). Distinct marker shapes and line styles provide visual encoding for greyscale readability.

TABLE V
NEGATIVE-FRAME RATIO SWEEP ACROSS ARCHITECTURES (RQ1)

Det.	r	mAP ₅₀	mAP _{50:95}	P	R	FP/1k
YOLO11n	0%	0.9824	0.7570	0.8576	0.9787	67.22
	20%	0.9838	0.7524	0.9283	0.9814	31.05
	40%	0.9919	0.7711	0.9308	0.9831	22.01
	60%	0.9899	0.7723	0.9468	0.9806	9.83
	80%	0.9933	0.7606	0.9644	0.9766	4.72
YOLO26n	0%	0.9170	0.7086	0.8557	0.9105	80.59
	20%	0.9729	0.7546	0.9306	0.9763	16.90
	40%	0.9726	0.7388	0.9224	0.9519	9.43
	60%	0.9660	0.7594	0.9587	0.9086	10.22
	80%	0.9736	0.7671	0.9217	0.9794	3.14
YOLO12n	0%	0.9910	0.7496	0.9228	0.9823	45.20
	20%	0.9757	0.7449	0.8954	0.9863	35.77
	40%	0.9866	0.7659	0.9199	0.9949	18.86
	60%	0.9888	0.7766	0.9348	0.9960	9.43
	80%	0.9925	0.7529	0.9621	0.9767	7.08
YOLOv10n	0%	0.9423	0.7287	0.9065	0.8831	57.39
	20%	0.9423	0.7270	0.8833	0.9400	21.23
	40%	0.9642	0.7568	0.8813	0.9788	14.94
	60%	0.9749	0.7525	0.9221	0.9758	4.72
	80%	0.9524	0.7384	0.9010	0.9024	4.72

Dual-seed means (seed $\in \{42, 43\}$) for $r \leq 40\%$; single seed (seed = 42) otherwise. FP/1k is computed at the fixed operational threshold $\tau = 0.25$ (NMS IoU 0.70) over 1,272 negative test frames. Precision (P) and recall (R) are reported at the confidence threshold that maximizes F_1 on the validation split for each configuration, which may differ from $\tau = 0.25$; the two are therefore two distinct operating points and should not be read as arising from the same confusion matrix. Bold indicates best performance per architecture.

five ratios (a repeated-measures structure); the reported p -values should be read as descriptive rather than as formal hypothesis tests, as no correction for this clustering is applied.

Cue Sensitivity and Localization Stability: Precision improves significantly with negative ratio (Spearman $\rho = +0.546, p < 0.05$) due directly to false-alarm suppression. In contrast, recall remains uncorrelated with negative ratio (Spearman $\rho = +0.012, p > 0.50$), confirming that negative frames do not cause under-detection. Localization accuracy (mAP_{50:95}) remains within a narrow operational band across all configurations (0.7086–0.7766), showing no systematic degradation with increasing negative exposure.

Intermediate Trajectory Differences: Although all models

converge at 80% negatives, their trajectories differ between 20% and 60%. YOLO26n drops sharply to 16.90 FP/1k at 20% negatives, plateauing early. In contrast, YOLO11n and YOLO12n show a more gradual decline requiring ratios between 60% and 80%, whereas YOLOv10n achieves early suppression by 40% (14.94 FP/1k) before reaching its 4.72 FP/1k floor at 60%.

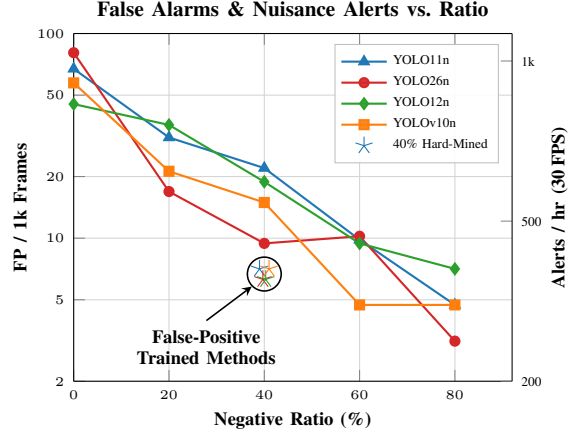


Fig. 5. Measured false-positive rates (left axis) and projected hourly alerts at 30 FPS (right axis; derived via (6)) versus negative ratio. Both metrics share synchronized logarithmic scales due to linear scaling ($\mathcal{A}_h = 87.4 \times \text{FP/1k}$). Circled cluster highlights 40% hard-negative mining results; star markers are horizontally jittered (39.0–41.0) purely for visual separation, all four represent the identical 40% ratio.

B. Hard-Negative Mining vs. Random Negatives

To address RQ2, we evaluate whether hard negatives improve suppression compared to random background frames at matched dataset volume. Following Section III-E, we harvest false positives from the 0% baseline detector ($\tau = 0.25$) and backfill with random negatives to reach 1,600 negative frames (4,001 total frames, matching the 40% ratio). Fig. 6 presents two-seed sample means (seed $\in \{42, 43\}$) comparing random negative sampling with hard-negative mining.

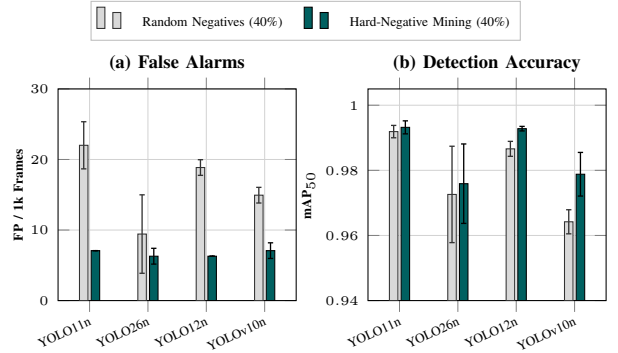


Fig. 6. Hard-negative mining versus random negative sampling ($r_{\text{ref}} = 40\%$, RQ2): (a) false alarms per 1,000 frames (FP/1k); (b) core detection accuracy (mAP₅₀). Bars report dual-seed sample means with ± 1 standard deviation error bars.

TABLE VI
PROJECTED NUISANCE ALERTS PER HOUR ($\mathcal{A}_h$) ACROSS NEGATIVE
RATIOS AND FRAME RATES

Model	Training Configuration	5 FPS	15 FPS	30 FPS
YOLO11n	0% Baseline	979	2,937	5,873
	40% Random	321	962	1,923
	40% Hard-Mined	103	309	619
	80% Ratio	69	206	412
YOLO26n	0% Baseline	1,174	3,521	7,041
	40% Random	137	412	824
	40% Hard-Mined	92	275	550
	80% Ratio	46	137	274
YOLO12n	0% Baseline	658	1,975	3,949
	40% Random	275	824	1,648
	40% Hard-Mined	92	275	550
	80% Ratio	103	309	619
YOLOv10n	0% Baseline	836	2,507	5,014
	40% Random	218	653	1,305
	40% Hard-Mined	103	309	619
	80% Ratio	69	206	412

Projected via (6) with $p_{\text{neg}} = 0.809$. FP/1k inputs correspond to Fig. 4 and Fig. 6. Bold indicates the lowest alert rate per architecture.

Architectural Variation in Suppression: Hard mining reduces false alarms across all four architectures, but the magnitude depends on model design (Fig. 6):

- *Attention and Dual-Assignment Models:* YOLO11n, YOLO12n, and YOLOv10n achieve substantial reductions: -67.8% for YOLO11n ($22.01 \rightarrow 7.08$ FP/1k), -66.7% for YOLO12n ($18.86 \rightarrow 6.29$ FP/1k), and -52.6% for YOLOv10n ($14.94 \rightarrow 7.08$ FP/1k). Localization quality also improves, with $\text{mAP}_{50:95}$ increasing by up to $+0.018$ in YOLO12n.
- *Purely Convolutional Models:* YOLO26n shows a smaller reduction of -33.3% ($9.43 \rightarrow 6.29$ FP/1k), demonstrating diminishing returns from targeted mining over random background frames.

We note that with dual-seed training ($n = 2$), these comparisons reflect empirical sample mean differences rather than formal hypothesis tests; given the variance observed in some configurations (e.g., YOLO26n random-40% standard deviation of ± 5.56 FP/1k), larger multi-seed batteries are required to establish statistical significance.

C. Nuisance Alert Projection

To evaluate operational reliability, we project measured false-alarm rates into expected hourly nuisance alerts ($\mathcal{A}_h$) using (6). While on-road naturalistic driving research indicates that potentially distracting secondary-task engagement occupies only a minority of total driving time [18], we evaluate alert projections using our held-out test stream prevalence of $p_{\text{neg}} = 0.809$ (80.9% negative frames) to maintain empirical consistency with our test partition. Table VI reports projected alert rates across three streaming rates: 5 FPS (low-power sampling), 15 FPS (standard embedded vision), and 30 FPS (full real-time streaming).

Alert Fatigue and Automation Disuse: Models trained without negative frames (0%) trigger impractically high alert

frequencies (Table VI). At 30 FPS, unregularized detectors trigger 3,949 to 7,041 alerts/hr (1–2 false alerts per second). Even at 5 FPS, they generate 658 to 1,174 alerts/hr. In human-automation interaction theory, this alert frequency exceeds driver tolerance, directly inducing alert fatigue and automation disuse [5], [13].

Cascading Edge Compute and Thermal Load: Continuous false alarms persistently trigger downstream facial tracking, logging, and video encoding routines. The resulting processing spikes can induce thermal throttling and reduce edge throughput by over 40% [3], [6].

Mitigation via Negative Curation: Curating negative training frames lowers projected alert rates by up to 96.1%. At an 80% ratio, alerts drop to 274–619 alerts/hr at 30 FPS and 46–103 alerts/hr at 5 FPS. Furthermore, hard-negative mining provides comparable suppression with one-third the dataset volume and one-sixth the negative frames: at 40%, hard-mined models produce 550–619 alerts/hr at 30 FPS, an 86% to 92% reduction from the 0% baseline depending on architecture.

These projections highlight an engineering distinction: *inference latency depends on architecture, whereas operational reliability depends on data curation*. While all four models execute within real-time limits (7.7–21.7 ms; Fig. 3), negative curation determines whether a detector avoids alert fatigue and thermal throttling.

V. DISCUSSION

A. Architectural Mechanisms and Practical Curation Guidelines

A detector’s feature-aggregation mechanism determines how it responds to negative-frame curation.

Mechanistic Divergence: Random negative sampling primarily selects low-entropy background patches that generate minimal training gradients. Purely convolutional networks, such as YOLO26n, rely on localized 3×3 re-parameterized convolutions (RepConv). These layers quickly learn repetitive cabin textures, causing YOLO26n to plateau at a 20% negative ratio and gain modest benefit from hard-negative mining (-33.3%). In contrast, models using self-attention (YOLO11n), area attention (YOLO12n), or dual-label assignment (YOLOv10n) capture wider spatial context. Without negative frames, attention layers often mistake cabin lighting or seat contours for driver cues. Mined hard negatives provide high-entropy distractors that force attention heads to distinguish cabin structures from driver actions. For YOLOv10n, mined negatives also penalize over-assignment in the auxiliary head, sharpening the decision boundary of the primary inference branch.

Per-Seed Stability: In our two-seed evaluations, attention-based and dual-assignment models showed consistent suppression across runs, whereas YOLO26n exhibited marked variability (ranging from 0.0% under Seed 42 to 47.0% under Seed 43). While this suggests potential seed sensitivity in localized convolutional backbones, evaluations across a larger battery of seeds are required before drawing definitive architectural conclusions.

Practical Curation Guidelines: These empirical results establish two engineering guidelines for edge driver monitoring systems (DMS):

- 1) *Backbone-Targeted Curation:* Uniform random sampling provides sufficient false-alarm suppression for convolutional backbones. However, attention-centric and dual-assignment models require hard-negative mining, yielding an additional 52.6% to 67.8% false-alarm reduction at matched training compute.
- 2) *Prevalence Trade-Offs (40% vs. 80%):* An 80% negative ratio achieves the lowest false-alarm floor for most architectures (3.14–4.72 FP/1k), though YOLO12n reaches its minimum under 40% hard mining (6.29 FP/1k). Where storage or compute is constrained, a 40% hard-mined ratio approaches 80% random suppression (6.29–7.08 FP/1k) while requiring only one-third the total dataset volume and one-sixth the negative frames.

B. Limitations

This evaluation has six primary limitations:

Single-Annotator Labeling: A single annotator labeled the visual cues without an independent review, which may introduce subjective bias in ambiguous edge cases.

Simplified Cue Annotation and Mutual Exclusion: Annotations enforced a single bounding box per positive frame and a strict mutual-exclusion rule between overlapping behaviors (e.g., covering a yawn is labeled exclusively as hand-over-mouth). In naturalistic driving, concurrent and overlapping cues frequently co-occur; this protocol simplifies the detection objective and may understate real-world false negatives when simultaneous behaviors arise.

Single-Modality RGB Sensing: The evaluation relies entirely on visible-light RGB video feeds without near-infrared (NIR) or active depth sensing commonly deployed for in-cabin night vision.

Isolated Frame-Level Inference: Detections were evaluated per frame without temporal persistence filtering (e.g., Kalman tracking or sliding time-windows), yielding an unmitigated upper-bound alert baseline.

Driver Partition Overlap and Single-Pass Mining: Splits were stratified across cue instances rather than subject-disjoint driver identities. Because frames were extracted at 1 FPS from continuous recordings, near-duplicate background frames from the same driver and session may appear in both a training split and a held-out split; this could inflate the reported false-alarm suppression relative to a fully subject-disjoint evaluation and should be treated as an upper bound on generalization until a driver-held-out replication is run. Hard-negative mining was also restricted to a single closed-loop iteration rather than multi-round continuous bootstrapping.

Desktop-GPU Deployment Proxy: Latency and memory profiling (Section III-F) was performed on a desktop-class discrete GPU rather than an automotive-grade embedded SoC, and no power draw was measured. Absolute latency, memory, and thermal margins on production in-cabin hardware may differ from the desktop-proxy figures reported here.

VI. CONCLUSION

This study evaluated negative-frame curation and hard-negative mining to suppress false alarms in edge-deployed Driver Monitoring Systems (DMS). Across four You Only Look Once (YOLO) architectures with fewer than 3 million parameters, increasing negative training frames from 0% to 80% reduced false positives per 1,000 frames (FP/1k) by $6.4\times$ to $25.7\times$. Core detection sensitivity ($\text{mAP}_{50:95}$) and recall remained stable while precision increased. At matched dataset volume, replacing random negative frames with hard-mined false positives achieved an additional 33.3% to 67.8% false-alarm reduction. Attention-based and dual-assignment models benefited most from hard negatives, whereas purely convolutional models achieved effective suppression using random background frames alone. In streaming projections, these curation protocols lowered nuisance alerts below 620 alerts/hr at 30 frames per second (FPS), mitigating driver alert fatigue and edge computational load.

These findings show that dataset curation is as critical as architectural design for edge deployment. Furthermore, curation strategies must match the detector’s internal feature-aggregation mechanism: attention heads require high-entropy mined distractors to eliminate false correlations, whereas convolutional kernels regularize effectively on uniform background scenes.

This study has six primary shortcomings: visual cues were labeled by a single annotator; annotations enforced single-cue mutual exclusion rather than simultaneous multi-cue interactions; evaluations relied exclusively on single-frame visible-light RGB feeds without temporal smoothing; mining was limited to a single pass; data splits were stratified across cue instances rather than disjoint drivers; and latency/memory profiling relied on a desktop-GPU proxy rather than validated automotive SoC hardware, with no power measurement. Future work should validate these curation protocols on multi-modal near-infrared (NIR) and depth sensors, incorporate multi-frame temporal consensus algorithms, evaluate iterative multi-round mining, test cross-subject generalization on disjoint driver splits, and repeat the latency and power-draw benchmarks on representative automotive SoC hardware.

REPRODUCIBILITY STATEMENT

To facilitate independent verification, training configuration scripts, hyperparameter files, and exact frame split indices will be made publicly available in an open-source archive upon publication.

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